

Vector Analysis

Problem 1.1

Show that the dot product and cross product is distributive

Solution:

Let the three vectors be

$$\mathbf{v}_1 = [a, b, c]$$

$$\mathbf{v}_2 = [d, e, f]$$

$$\mathbf{v}_3 = [g, h, i]$$

dot product:

Show

$$\mathbf{v}_1 \cdot (\mathbf{v}_2 + \mathbf{v}_3) = \mathbf{v}_1 \cdot \mathbf{v}_2 + \mathbf{v}_1 \cdot \mathbf{v}_3$$

Using

$$\mathbf{v}_2 + \mathbf{v}_3 = [d + g, e + h, f + i]$$

$$\mathbf{v}_1 \cdot \mathbf{v}_2 = ad + be + cf$$

$$\mathbf{v}_1 \cdot \mathbf{v}_3 = ag + bh + ci$$

LHS:

$$\begin{aligned} \mathbf{v}_1 \cdot (\mathbf{v}_2 + \mathbf{v}_3) &= [a, b, c] \cdot [d + g, e + h, f + i] = \\ &= a(d + g) + b(e + h) + c(f + i) \end{aligned}$$

RHS:

$$\begin{aligned} \mathbf{v}_1 \cdot \mathbf{v}_2 + \mathbf{v}_1 \cdot \mathbf{v}_3 &= ad + be + cf + ag + bh + ci = \\ &= a(d + g) + b(e + h) + c(f + i) \end{aligned}$$

cross product: Show

$$\mathbf{v}_1 \times (\mathbf{v}_2 + \mathbf{v}_3) = (\mathbf{v}_1 \times \mathbf{v}_2) + (\mathbf{v}_1 \times \mathbf{v}_3)$$

Using

$$\begin{aligned} \mathbf{v}_2 + \mathbf{v}_3 &= [d + g, e + h, f + i] \\ \mathbf{v}_1 \times \mathbf{v}_2 &= bf - ce + cd - af + ae - bd \\ \mathbf{v}_1 \times \mathbf{v}_3 &= bi - ch + cg - ai + ah - bg \end{aligned}$$

which gives

$$(\mathbf{v}_1 \times \mathbf{v}_2) + (\mathbf{v}_1 \times \mathbf{v}_3) = a(e + h - f - i) + b(f + i - d - g) + c(d + g - e - h)$$

and LHS:

$$\begin{aligned} \mathbf{v}_1 \times (\mathbf{v}_2 + \mathbf{v}_3) &= b(f + i) - c(e + h) + c(d + g) - a(f + i) + a(e + h) - b(d + g) = \\ &= bf + bi - ce - ch + cd + cg - af - ai + ae + ah - bd - bg = \\ &= a(e + h - f - i) + b(f + i - d - g) + c(d + g - e - h) \end{aligned}$$

□

Problem 1.2

Is the cross product associative?

$$(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} \stackrel{?}{=} \mathbf{A} \times (\mathbf{B} \times \mathbf{C})$$

Solution:

No, for example set

$$\begin{aligned} \mathbf{A} &= \hat{x} \\ \mathbf{B} &= \hat{x} \\ \mathbf{C} &= \hat{z} \end{aligned}$$

Then, the left hand side above becomes

$$(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} = (\hat{x} \times \hat{x}) \times \hat{z} = 0$$

while the right hand side is

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \hat{x} \times (\hat{x} \times \hat{z}) = \hat{x} \times -\hat{y} = -\hat{z}$$

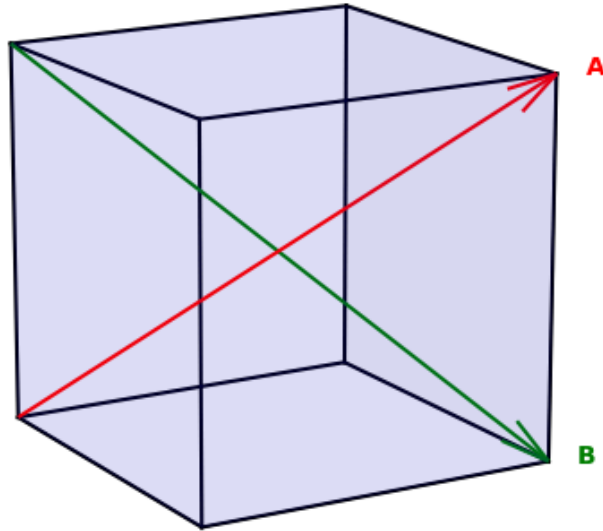
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Problem 1.3

Find the angle between the body diagonals of a cube.

Solution:

The body diagonals $\mathbf{A} = [1, 1, 1]$ and $\mathbf{B} = [1, 1, -1]$



$$|\mathbf{A}| = |\mathbf{B}| = \sqrt{3}$$

$$\mathbf{A} \cdot \mathbf{B} = 1 + 1 - 1 = 1$$

$$\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}| |\mathbf{B}| \cos \theta = 3 \cos \theta = 1 \rightarrow \theta = \arccos\left(\frac{1}{3}\right) \approx 70.53^\circ \quad \square$$

Problem 1.4

Use the cross product to find the components of the unit vector $\hat{\mathbf{n}}$ perpendicular to the shaded plane in Fig.

Solution:

Two of the plane vectors are $\mathbf{v}_1 = [-1, 2, 0]$ and $\mathbf{v}_2 = [-1, 0, 3]$. $\mathbf{v}_1 \times \mathbf{v}_2 = \mathbf{n} = [6, 3, 2]$ and $|\mathbf{n}| = 7$ gives $\hat{\mathbf{n}} = \frac{1}{7}[6, 3, 2] \quad \square$

Problem 1.5

Prove the **BAC-CAB** rule,

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$$

by writing out both sides in component form.

Solution:

Writing out the vectors in component form

$$\mathbf{A} = A_x \hat{\mathbf{x}} + A_y \hat{\mathbf{y}} + A_z \hat{\mathbf{z}}$$

$$\mathbf{B} = B_x \hat{\mathbf{x}} + B_y \hat{\mathbf{y}} + B_z \hat{\mathbf{z}}$$

$$\mathbf{C} = C_x \hat{\mathbf{x}} + C_y \hat{\mathbf{y}} + C_z \hat{\mathbf{z}}$$

starting with lhs

$$\mathbf{B} \times \mathbf{C} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix} = (B_y C_z - B_z C_y) \hat{\mathbf{x}} + (B_z C_x - B_x C_z) \hat{\mathbf{y}} + (B_x C_y - B_y C_x) \hat{\mathbf{z}}$$

rewriting with a new matrix $\mathbf{D}$ where

$$D_x = B_y C_z - B_z C_y$$

$$D_y = B_z C_x - B_x C_z$$

$$D_z = B_x C_y - B_y C_x$$

gives

$$\mathbf{D} = \mathbf{B} \times \mathbf{C} = D_x \hat{\mathbf{x}} + D_y \hat{\mathbf{y}} + D_z \hat{\mathbf{z}}$$

Now

$$\begin{aligned} \mathbf{A} \times (\mathbf{B} \times \mathbf{C}) &= \mathbf{A} \times \mathbf{D} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ A_x & A_y & A_z \\ D_x & D_y & D_z \end{vmatrix} = \\ &= (A_y D_z - A_z D_y) \hat{\mathbf{x}} + (A_z D_x - A_x D_z) \hat{\mathbf{y}} + (A_x D_y - A_y D_x) \hat{\mathbf{z}} \end{aligned}$$

RHS:

$$\mathbf{A} \cdot \mathbf{C} = A_x C_x + A_y C_y + A_z C_z = P$$

$$\mathbf{A} \cdot \mathbf{B} = A_x B_x + A_y B_y + A_z B_z = Q$$

$$\mathbf{B}(\mathbf{A} \cdot \mathbf{C}) = B_x P \hat{\mathbf{x}} + B_y P \hat{\mathbf{y}} + B_z P \hat{\mathbf{z}}$$

$$\mathbf{C}(\mathbf{A} \cdot \mathbf{B}) = C_x Q \hat{\mathbf{x}} + C_y Q \hat{\mathbf{y}} + C_z Q \hat{\mathbf{z}}$$

$$\mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}) = (B_x P - C_x Q) \hat{\mathbf{x}} + (B_y P - C_y Q) \hat{\mathbf{y}} + (B_z P - C_z Q) \hat{\mathbf{z}}$$

Now final step is checking if

$$\mathbf{1:} \quad A_y D_z - A_z D_y = B_x P - C_x Q$$

$$\mathbf{2:} \quad A_z D_x - A_x D_z = B_y P - C_y Q$$

$$\mathbf{3:} \quad A_x D_y - A_y D_x = B_z P - C_z Q$$

For **1**:

$$\begin{aligned} \text{LHS: } & A_y(B_x C_y - B_y C_x) - A_z(B_z C_x - B_x C_z) \\ &= A_y B_x C_y - A_y B_y C_x - A_z B_z C_x + A_z B_x C_z \\ &= B_x(A_y C_y + A_z C_z) - C_x(A_y B_y + A_z B_z) \end{aligned}$$

$$\begin{aligned} \text{RHS: } & B_x(A_x C_x + A_y C_y + A_z C_z) - C_x(A_x B_x + A_y B_y + A_z B_z) \\ &= B_x A_x C_x + B_x(A_y C_y + A_z C_z) - C_x A_x B_x - C_x(A_y B_y + A_z B_z) \\ &= B_x(A_y C_y + A_z C_z) - C_x(A_y B_y + A_z B_z) = \text{LHS } \checkmark \end{aligned}$$

For **2**:

$$\begin{aligned} \text{LHS: } & A_z(B_y C_z - B_z C_y) - A_x(B_x C_y - B_y C_x) \\ &= A_z B_y C_z - A_z B_z C_y - A_x B_x C_y + A_x B_y C_x \\ &= B_y(A_z C_z + A_x C_x) - C_y(A_z B_z + A_x B_x) \end{aligned}$$

$$\begin{aligned} \text{RHS: } & B_y(A_x C_x + A_y C_y + A_z C_z) - C_y(A_x B_x + A_y B_y + A_z B_z) \\ &= B_y A_x C_x + B_y(A_y C_y + A_z C_z) - C_y A_x B_x - C_y(A_y B_y + A_z B_z) \\ &= B_y(A_x C_x + A_z C_z) - C_y(A_x B_x + A_z B_z) = \text{LHS } \checkmark \end{aligned}$$

For **3**:

$$\begin{aligned} \text{LHS: } & A_x(B_z C_x - B_x C_z) - A_y(B_y C_z - B_z C_y) \\ &= A_x B_z C_x - A_x B_x C_z - A_y B_y C_z + A_y B_z C_y \\ &= B_z(A_x C_x + A_y C_y) - C_z(A_x B_x + A_y B_y) \end{aligned}$$

$$\begin{aligned} \text{RHS: } & B_z(A_x C_x + A_y C_y + A_z C_z) - C_z(A_x B_x + A_y B_y + A_z B_z) \\ &= B_z A_x C_x + B_z(A_y C_y + A_z C_z) - C_z A_x B_x - C_z(A_y B_y + A_z B_z) \\ &= B_z(A_x C_x + A_y C_y) - C_z(A_x B_x + A_y B_y) = \text{LHS } \checkmark \end{aligned}$$

Since all three components are equal, the BAC-CAB rule is proven. $\square$

Problem 1.6

Prove that

$$[\mathbf{A} \times (\mathbf{B} \times \mathbf{C})] + [\mathbf{B} \times (\mathbf{C} \times \mathbf{A})] + [\mathbf{C} \times (\mathbf{A} \times \mathbf{B})] = \mathbf{0}$$

Under what conditions does $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \times \mathbf{B}) \times \mathbf{C}$?

Solution:

Using the BAC-CAB rule, rewrite the triple cross product as

$$(1) \mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$$

$$(2) \mathbf{B} \times (\mathbf{C} \times \mathbf{A}) = \mathbf{C}(\mathbf{B} \cdot \mathbf{A}) - \mathbf{A}(\mathbf{B} \cdot \mathbf{C})$$

$$(3) \mathbf{C} \times (\mathbf{A} \times \mathbf{B}) = \mathbf{A}(\mathbf{C} \cdot \mathbf{B}) - \mathbf{B}(\mathbf{C} \cdot \mathbf{A})$$

Using the commutative property of the dot product, $\mathbf{v}_1 \cdot \mathbf{v}_2 = \mathbf{v}_2 \cdot \mathbf{v}_1$, it is seen that:

- The $\mathbf{B}$ -part in (1) cancels out the $\mathbf{B}$ -part in (3)
- The $\mathbf{C}$ -part in (1) cancels out the $\mathbf{C}$ -part in (2)
- The $\mathbf{A}$ -part in (2) cancels out the $\mathbf{A}$ -part in (3)

The conditions when $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \times \mathbf{B}) \times \mathbf{C}$ is translated with the BAC-CAB rule into $\mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}) = -\mathbf{A}(\mathbf{C} \cdot \mathbf{B}) + \mathbf{B}(\mathbf{C} \cdot \mathbf{A})$ which gives that either $\mathbf{A} = \mathbf{C}$ or $\mathbf{B} \perp (\mathbf{A} \text{ and } \mathbf{C})$ $\square$

Problem 1.7

Find the separation vector $\mathbf{r}$ from the source point $(2, 8, 7)$ to the field point $(4, 6, 8)$. Determine its magnitude r , and construct the unit vector $\hat{\mathbf{r}}$.

Solution:

The separation vector $\mathbf{r}$ is the difference between the field point and the source point.

$$\mathbf{r} = (4, 6, 8) - (2, 8, 7) = (2, -2, 1)$$

The magnitude is

$$|\mathbf{r}| = r = \sqrt{2^2 + (-2)^2 + 1^2} = 3$$

and the unit vector is

$$\frac{\mathbf{r}}{r} = \hat{\mathbf{r}} = \frac{1}{3}(2, -2, 1)$$

$\square$

Problem 1.8**Problem 1.9****Problem 1.10****Problem 1.11**

Find the gradients of the following functions:

- (a) $f(x, y, z) = x^2 + y^3 + z^4$
- (b) $f(x, y, z) = x^2 y^3 z^4$
- (c) $f(x, y, z) = e^x \sin(y) \ln(z)$

Solution:

The gradient of a function T is defined as

$$\nabla T = \frac{\partial T}{\partial x} \hat{x} + \frac{\partial T}{\partial y} \hat{y} + \frac{\partial T}{\partial z} \hat{z}$$

This results in

- (a) $2x\hat{x} + 3y^2\hat{y} + 4z^3\hat{z}$
- (b) $2xy^3z^4\hat{x} + 3x^2y^2z^4\hat{y} + 4x^2y^3z^3\hat{z}$
- (c) $e^x \left(\sin(y) \ln(z) \hat{x} + \cos(y) \ln(z) \hat{y} + \frac{\sin(y)}{z} \hat{z} \right)$

□

Problem 1.12

The height of a certain hill (in feet) is given by

$$h(x, y) = 10(2xy - 3x^2 - 4y^2 - 18x + 28y + 12)$$

where y is the distance (in miles) north, x the distance east, of South Hadley.

- (a) Where is the top of the hill located?
- (b) How high is the hill?

(c) How steep is the slope (in feet per mile) at a point 1 mile north and 1 mile east of South Hadley? In what direction is the slope steepest, at that point?

Solution:

The gradient of $h(x, y)$ is

$$\nabla h = (20y - 60x - 180)\hat{x} + (20x - 80y + 280)\hat{y}$$

(a) The location of the top is calculated by setting the gradient to 0. This means

$$20y - 60x - 180 = 0$$

$$20x - 80y + 280 = 0$$

which has the solution $(x, y) = (-2, 3)$ miles.

(b) Putting the values given by (a) into the height function $h(x, y)$, the height of the hill is $h(-2, 3) = 720$ ft.

(c) At the point $(x, y) = (1, 1)$, the steepness and direction is obtained by putting this point into the gradient. The slope is then $\sqrt{2} \cdot 220$ ft/mile in the direction $-\hat{x} + \hat{y}$, which corresponds to northwest. $\square$

Problem 1.13

Let $\mathbf{r}$ be the separation vector from a fixed point (x', y', z') to the point (x, y, z) , and let r be its length. Show that

(a) $\nabla(r^2) = 2 \mathbf{r}$

(b) $\nabla\left(\frac{1}{r}\right) = -\hat{\mathbf{r}}/r^2$

(c) What is the general formula for $\nabla(r^n)$?

Solution:

$$\mathbf{r} = (x - x', y - y', z - z')$$

$$r = \sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2}$$

(a)

$$\begin{aligned}\nabla(r^2) &= \nabla\left((x - x')^2 + (y - y')^2 + (z - z')^2\right) = \\ &= 2(x - x')\hat{x} + 2(y - y')\hat{y} + 2(z - z')\hat{z} = 2 \mathbf{r}\end{aligned}$$

(b)

$$\begin{aligned}\nabla\left(\frac{1}{r}\right) &= \nabla\left(\frac{1}{\sqrt{(x-x')^2 + (y-y')^2 + (z-z')^2}}\right) = \\ &= -\frac{1}{2} \frac{2(x-x')\hat{x} + 2(y-y')\hat{y} + 2(z-z')\hat{z}}{\sqrt{(x-x')^2 + (y-y')^2 + (z-z')^2}^3} = -\frac{\mathbf{r}}{r^3} = -\frac{\hat{\mathbf{r}}}{r^2}\end{aligned}$$

(c)

$$\begin{aligned}\nabla(r^n) &= \nabla\left(\left((x-x')^2 + (y-y')^2 + (z-z')^2\right)^{\frac{n}{2}}\right) = \\ &= \frac{n}{2}(2(x-x')\hat{x} + 2(y-y')\hat{y} + 2(z-z')\hat{z})\left[(x-x')^2 + (y-y')^2 + (z-z')^2\right]^{\frac{n-2}{2}} = \\ &= nr^{n-2}\hat{\mathbf{r}}\end{aligned}$$

□

Problem 1.14

Problem 1.15

Calculate the divergence of the following vector functions:

(a) $\mathbf{v}_a = x^2\hat{x} + 3xz^2\hat{y} - 2xz\hat{z}$

(b) $\mathbf{v}_b = xy\hat{x} + 2yz\hat{y} + 3xz\hat{z}$

(c) $\mathbf{v}_c = y^2\hat{x} + (2xy + z^2)\hat{y} + 2yz\hat{z}$

Solution:

The divergence of a vector $\mathbf{v}$ is defined as

$$\nabla \cdot \mathbf{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}$$

(a) $\nabla \cdot \mathbf{v}_a = 2x - 2x = 0$

(b) $\nabla \cdot \mathbf{v}_b = y + 2z + 3x$

(c) $\nabla \cdot \mathbf{v}_c = 2x + 2y$

□

Problem 1.16

Sketch the vector function

$$\mathbf{v} = \frac{\hat{\mathbf{r}}}{r^2},$$

and compute its divergence.

Solution:

The divergence of the r-component in spherical coordinates is

$$\nabla \cdot \mathbf{v} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 v_r)$$

Using this, the divergence is $\nabla \cdot \mathbf{v} = 0$. From the sketch, clearly the divergence is not 0 at the origin. The problem arises at the point $\mathbf{r} = 0$ where $\mathbf{v} \rightarrow \infty$. At all other points $\mathbf{r} \neq 0$, the divergence is 0.

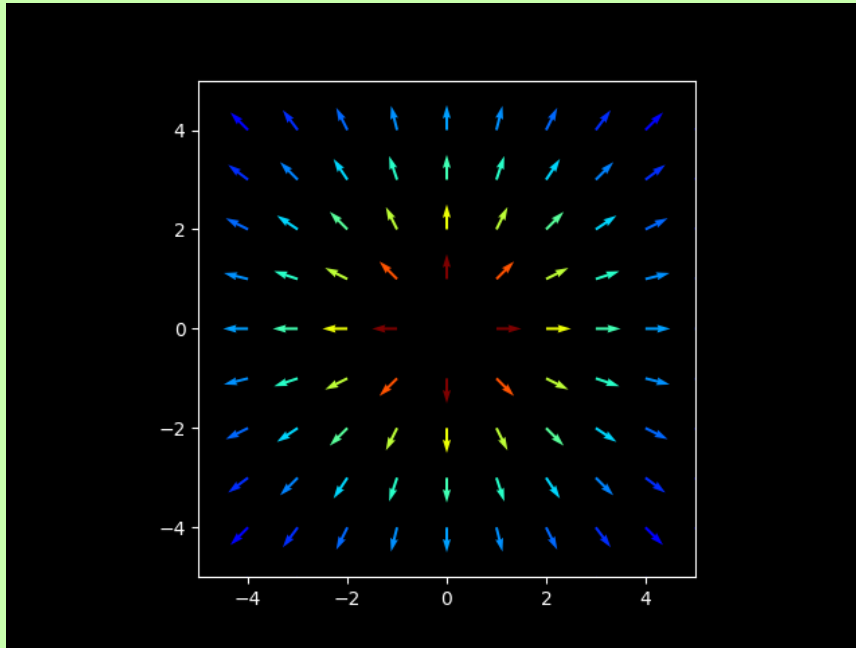


Figure 2: Sketch of the function $\mathbf{v} = \frac{\hat{\mathbf{r}}}{r^2}$

□

Problem 1.17**Problem 1.18**

Calculate the curls of the vector functions

(a) $\mathbf{v}_a = x^2\hat{\mathbf{x}} + 3xz^2\hat{\mathbf{y}} - 2xz\hat{\mathbf{z}}$

(b) $\mathbf{v}_b = xy\hat{\mathbf{x}} + 2yz\hat{\mathbf{y}} + 3xz\hat{\mathbf{z}}$

(c) $\mathbf{v}_c = y^2\hat{\mathbf{x}} + (2xy + z^2)\hat{\mathbf{y}} + 2yz\hat{\mathbf{z}}$

Solution:

The definition of the curl of a vector function $\mathbf{v}$ is

$$\nabla \times \mathbf{v} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_x & v_y & v_z \end{vmatrix} = \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) \hat{\mathbf{x}} + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) \hat{\mathbf{y}} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) \hat{\mathbf{z}}$$

This gives

(a) $\nabla \times \mathbf{v}_a = -6xz\hat{\mathbf{x}} + 2z\hat{\mathbf{y}} + 3z^2\hat{\mathbf{z}}$

(b) $\nabla \times \mathbf{v}_b = -2y\hat{\mathbf{x}} - 3z\hat{\mathbf{y}} - x\hat{\mathbf{z}}$

(c) $\nabla \times \mathbf{v}_c = 0$

□

Problem 1.19

Problem 1.20

Construct a vector function that has a zero divergence and zero curl everywhere.

Solution:

The vector function $\mathbf{R}$ should satisfy

$$\nabla \cdot \mathbf{R} = 0$$

$$\nabla \times \mathbf{R} = 0$$

Using the definitions of the divergence and curl, the conditions are

$$\nabla \cdot \mathbf{R} = \frac{\partial R_x}{\partial x} + \frac{\partial R_y}{\partial y} + \frac{\partial R_z}{\partial z} = 0$$

$$\nabla \times \mathbf{R} = \left(\frac{\partial R_z}{\partial y} - \frac{\partial R_y}{\partial z} \right) \hat{\mathbf{x}} + \left(\frac{\partial R_x}{\partial z} - \frac{\partial R_z}{\partial x} \right) \hat{\mathbf{y}} + \left(\frac{\partial R_y}{\partial x} - \frac{\partial R_x}{\partial y} \right) \hat{\mathbf{z}}$$

Want to fulfill:

$$\frac{\partial R_z}{\partial y} = \frac{\partial R_y}{\partial z}$$

$$\frac{\partial R_x}{\partial z} = \frac{\partial R_z}{\partial x}$$

$$\frac{\partial R_y}{\partial x} = \frac{\partial R_x}{\partial y}$$

An example is $\mathbf{R} = yz\hat{x} + xz\hat{y} + xy\hat{z}$

□

Problem 1.21

Prove the product rules

- (i) $\nabla(fg) = f\nabla g + g\nabla f$
- (iv) $\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B})$
- (v) $\nabla \times (f\mathbf{A}) = f(\nabla \times \mathbf{A}) - \mathbf{A} \times (\nabla f)$

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- (ii) $\nabla(\mathbf{A} \cdot \mathbf{B}) = \mathbf{A} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{A}) + (\mathbf{A} \cdot \nabla)\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{A}$
- (iv) $\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B})$
- (vi) $\nabla \times (\mathbf{A} \times \mathbf{B}) = (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B} + \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A})$

(a) Check product rule (iv) for the functions

$$\mathbf{A} = x\hat{x} + 2y\hat{y} + 3z\hat{z}, \quad \mathbf{B} = 3y\hat{x} - 2x\hat{y}$$

(b) Do the same for product rule (ii).

(c) Do the same for rule (vi).

Solution:

(a) Start with LHS:

$$\begin{aligned}
\nabla \cdot (\mathbf{A} \times \mathbf{B}) &= \nabla \cdot ((A_y B_z - A_z B_y)\hat{x} + (A_z B_x - A_x B_z)\hat{y} + (A_x B_y - A_y B_x)\hat{z}) = \\
&= \frac{\partial}{\partial x}(A_y B_z - A_z B_y) + \frac{\partial}{\partial y}(A_z B_x - A_x B_z) + \frac{\partial}{\partial z}(A_x B_y - A_y B_x) = \\
&= \frac{\partial}{\partial x}(6xz) + \frac{\partial}{\partial y}(9zy) + \frac{\partial}{\partial z}(-2x^2 - 6y^2) = 6z + 9z = 15z
\end{aligned}$$

RHS:

$$\begin{aligned}
&\underbrace{\mathbf{B} \cdot (\nabla \times \mathbf{A})}_{(1)} - \underbrace{\mathbf{A} \cdot (\nabla \times \mathbf{B})}_{(2)} = \\
(1) \quad &\mathbf{B} \cdot \left[\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \hat{x} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \hat{y} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \hat{z} \right] = \\
&= B_x \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + B_y \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + B_z \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) = 0 \\
(2) \quad &\mathbf{A} \cdot \left[\left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} \right) \hat{x} + \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} \right) \hat{y} + \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} \right) \hat{z} \right] = \\
&= A_x \left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} \right) + A_y \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} \right) + A_z \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} \right) = \\
&= 3z(-5) = -15z \\
&(1) - (2) = 0 - (-15z) = 15z
\end{aligned}$$

(b) LHS:

$$\begin{aligned}
\nabla(\mathbf{A} \cdot \mathbf{B}) &= \nabla(A_x B_x + A_y B_y + A_z B_z) = \nabla(3xy - 4xy) = \\
&= \frac{\partial(-xy)}{\partial x} \hat{x} + \frac{\partial(-xy)}{\partial y} \hat{y} = -y \hat{x} - x \hat{y}
\end{aligned}$$

RHS:

$$\underbrace{\mathbf{A} \times (\nabla \times \mathbf{B})}_{(1)} + \underbrace{\mathbf{B} \times (\nabla \times \mathbf{A})}_{(2)} + \underbrace{(\mathbf{A} \cdot \nabla) \mathbf{B}}_{(3)} + \underbrace{(\mathbf{B} \cdot \nabla) \mathbf{A}}_{(4)}$$

$$(1) \mathbf{A} \times \left[\left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} \right) \hat{x} + \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} \right) \hat{y} + \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} \right) \hat{z} \right] =$$

$$= (x\hat{x} + 2y\hat{y} + 3z\hat{z}) \times (-5\hat{z}) = -10y\hat{x} + 5x\hat{y}$$

$$(2) \mathbf{B} \times \left[\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \hat{x} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \hat{y} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \hat{z} \right] = 0$$

$$(3) (\mathbf{A} \cdot \nabla) \mathbf{B} =$$

$$= \left(A_x \frac{\partial B_x}{\partial x} + A_y \frac{\partial B_x}{\partial y} + A_z \frac{\partial B_x}{\partial z} \right) \hat{x}$$

$$+ \left(A_x \frac{\partial B_y}{\partial x} + A_y \frac{\partial B_y}{\partial y} + A_z \frac{\partial B_y}{\partial z} \right) \hat{y}$$

$$+ \left(A_x \frac{\partial B_z}{\partial x} + A_y \frac{\partial B_z}{\partial y} + A_z \frac{\partial B_z}{\partial z} \right) \hat{z} =$$

$$= 6y\hat{x} - 2x\hat{y}$$

$$(4) (\mathbf{B} \cdot \nabla) \mathbf{A} =$$

$$= \left(B_x \frac{\partial A_x}{\partial x} + B_y \frac{\partial A_x}{\partial y} + B_z \frac{\partial A_x}{\partial z} \right) \hat{x}$$

$$+ \left(B_x \frac{\partial A_y}{\partial x} + B_y \frac{\partial A_y}{\partial y} + B_z \frac{\partial A_y}{\partial z} \right) \hat{y}$$

$$+ \left(B_x \frac{\partial A_z}{\partial x} + B_y \frac{\partial A_z}{\partial y} + B_z \frac{\partial A_z}{\partial z} \right) \hat{z} =$$

$$= 3y\hat{x} - 4x\hat{y}$$

$$(1) + (2) + (3) + (4) = (-10y + 6y + 3y)\hat{x} + (5x - 2x - 4x)\hat{y} = -y\hat{x} - x\hat{y}$$

(c) LHS:

$$\nabla \times (\mathbf{A} \times \mathbf{B}) = \nabla \times [(A_y B_z - A_z B_y)\hat{x} + (A_z B_x - A_x B_z)\hat{y} + (A_x B_y - A_y B_x)\hat{z}] =$$

$$= \nabla \times [6xz\hat{x} + 9yz\hat{y} - (2x^2 + 6y^2)\hat{z}] = \nabla \times \mathbf{C} =$$

$$= \left(\frac{\partial C_z}{\partial y} - \frac{\partial C_y}{\partial z} \right) \hat{x} + \left(\frac{\partial C_x}{\partial z} - \frac{\partial C_z}{\partial x} \right) \hat{y} + \left(\frac{\partial C_y}{\partial x} - \frac{\partial C_x}{\partial y} \right) \hat{z} =$$

$$= (-12y - 9y)\hat{x} + (6x + 4x)\hat{y} = -21y\hat{x} + 10x\hat{y}$$

RHS: From (b) it was calculated that $(\mathbf{A} \cdot \nabla)\mathbf{B} = 6y\hat{x} - 2x\hat{y}$ and $(\mathbf{B} \cdot \nabla)\mathbf{A} = 3y\hat{x} - 4x\hat{y}$. The remaining parts of (vi) RHS is

$$\underbrace{\mathbf{A}(\nabla \cdot \mathbf{B})}_{=0} - \underbrace{\mathbf{B}(\nabla \cdot \mathbf{A})}_{=6} = -6\mathbf{B} = -18y\hat{x} + 12x\hat{y}$$

Summing it all up according to (vi), the result is

$$(3y - 6y - 18y)\hat{x} + (-4x + 2x + 12x)\hat{y} = -21y\hat{x} + 10x\hat{y}$$

□

Problem 1.26

Calculate the Laplacian of the following functions:

- (a) $T_a = x^2 + 2xy + 3z + 4$
- (b) $T_b = \sin x \sin y \sin z$
- (c) $T_c = e^{-5x} \sin 4y \cos 3z$
- (d) $\mathbf{v} = x^2\hat{x} + 3xz^2\hat{y} - 2xz\hat{z}$

Solution:

The Laplacian of a function T is defined as

$$\nabla^2 T = \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}$$

and the Laplacian of a vector $\mathbf{v}$ is defined as

$$\nabla^2 \mathbf{v} = (\nabla^2 v_x)\hat{x} + (\nabla^2 v_y)\hat{y} + (\nabla^2 v_z)\hat{z}$$

Using this, the result is

- (a) $\nabla^2 T_a = 2$
- (b) $\nabla^2 T_b = -3 \sin x \sin y \sin z$
- (c) $\nabla^2 T_c = e^{-5x} \sin 4y \cos 3z (25 - 16 - 9) = 0$
- (d) $\nabla^2 \mathbf{v} = 2\hat{x} + 6x\hat{y}$

□

Problem 1.27

Prove that the divergence of a curl is always zero.

Solution:

For a vector function $\mathbf{v}$, show

$$\nabla \cdot (\nabla \times \mathbf{v}) = 0$$

Using the definition of the divergence and curl:

$$\begin{aligned}\nabla \cdot \mathbf{v} &= \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = 0 \\ \nabla \times \mathbf{v} &= \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) \hat{\mathbf{x}} + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) \hat{\mathbf{y}} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) \hat{\mathbf{z}}\end{aligned}$$

the result is

$$\begin{aligned}\nabla \cdot (\nabla \times \mathbf{v}) &= \frac{\partial}{\partial x} \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) + \frac{\partial}{\partial y} \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) + \frac{\partial}{\partial z} \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) = \\ &= \frac{\partial^2 v_z}{\partial x \partial y} - \frac{\partial^2 v_y}{\partial x \partial z} + \frac{\partial^2 v_x}{\partial y \partial z} - \frac{\partial^2 v_z}{\partial y \partial x} + \frac{\partial^2 v_y}{\partial z \partial x} - \frac{\partial^2 v_x}{\partial z \partial y} = \\ &= \underbrace{\frac{\partial^2 v_z}{\partial x \partial y} - \frac{\partial^2 v_z}{\partial y \partial x}}_{=0} + \underbrace{\frac{\partial^2 v_x}{\partial y \partial z} - \frac{\partial^2 v_x}{\partial z \partial y}}_{=0} + \underbrace{\frac{\partial^2 v_y}{\partial z \partial x} - \frac{\partial^2 v_y}{\partial x \partial z}}_{=0}\end{aligned}$$

since the symmetry of second derivatives ensures that

$$\frac{\partial^2}{\partial x \partial y} = \frac{\partial^2}{\partial y \partial x}$$

□

Problem 1.28

Prove that the curl of a gradient is always zero.

Solution:

For a function T , show

$$\nabla \times (\nabla T) = 0$$

Using the definitions of the gradient and curl

$$\nabla T = \frac{\partial T}{\partial x} \hat{x} + \frac{\partial T}{\partial y} \hat{y} + \frac{\partial T}{\partial z} \hat{z}$$

$$\nabla \times \mathbf{v} = \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) \hat{x} + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) \hat{y} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) \hat{z}$$

results in

$$\nabla \times (\nabla T) = \underbrace{\left(\frac{\partial^2 T}{\partial y \partial z} - \frac{\partial^2 T}{\partial z \partial y} \right)}_{=0} \hat{x} + \underbrace{\left(\frac{\partial^2 T}{\partial z \partial x} - \frac{\partial^2 T}{\partial x \partial z} \right)}_{=0} \hat{y} + \underbrace{\left(\frac{\partial^2 T}{\partial x \partial y} - \frac{\partial^2 T}{\partial y \partial x} \right)}_{=0} \hat{z}$$

since the symmetry of second derivatives ensures that

$$\frac{\partial^2}{\partial x \partial y} = \frac{\partial^2}{\partial y \partial x}$$

□

Problem 1.29

Problem 1.30

Problem 1.31

Problem 1.32

Problem 1.33

Problem 1.34

Problem 1.35

Problem 1.36

Problem 1.37

Problem 1.38

Problem 1.39

Problem 1.40

Problem 1.41

Problem 1.42

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Problem 1.62

Problem 1.63

Problem 1.64

Problem 1.65